

Shift Register Micro-Architecture Document

(Enhanced)

1. Introduction

This document describes the micro-architecture of a unified shift-register module supporting SISO, SIPO, PISO, and PIPO operations. Additional signals for indicators are included.

2. High-Level Block Diagram

Below is a conceptual block diagram: +-----+ | SHIFT REGISTERS | | serial_in -> [SISO] -----> serial_out | | serial_in -> [SIPO] -----> parallel | | parallel -> [PISO] -----> serial_out | | parallel -> [PIPO] -----> parallel | +-----+

3. Valid Signals

- **serial_in_valid**: Indicates incoming serial bit is valid.
- **serial_out_valid**: Indicates output serial bit is valid.

These prevent unintended shifting and ensure synchronized data movement.

Signal	Direction	Description
clk	Input	Common clock for all submodules
rst_n	Input	Active-low reset
one_hot[0]	Input	Selects SISO mode
one_hot[1]	Input	Selects SIPO mode
one_hot[2]	Input	Selects PISO mode
one_hot[3]	Input	Selects PIPO mode
serial_in_valid	Input	Indicates valid serial input
serial_out_valid	Output	Indicates valid serial output bit

4. Datapath Description

SISO: A 1-bit flip-flop path shifting on each clock when valid.

SIPO: A 4-bit shift register loading serial bits LSB-first.

PISO: Parallel data is loaded into a 4-bit shift register and shifted out MSB-first.

PIPO: Simple parallel register capturing input on each clock when selected.